EVALUATION II

UNLIMITED SAMPLING-BASED HIGH-DYNAMIC-RANGE SPEECH ACQUISITION AND RECONSTRUCTION UNDER PRACTICAL ADC CONDITIONS

23ECE498 & 23EAC498 — PROJECT PHASE I

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THE ACOUSTIC DILEMMA: CLIPPING

Acoustic Clipping Bottleneck

- **Finite Rail Saturation:** Traditional ADCs irreversibly flatten amplitudes when $|x(t)| > \lambda$, permanently destroying spectral crests.
- **Harmonic Splatter:** Sharp rail transitions generate spurious high-frequency harmonics, distorting formant striations.
- **Downstream Failure:** Saturated signals degrade intelligibility, causing catastrophic word error rate (WER) penalties in ASR.

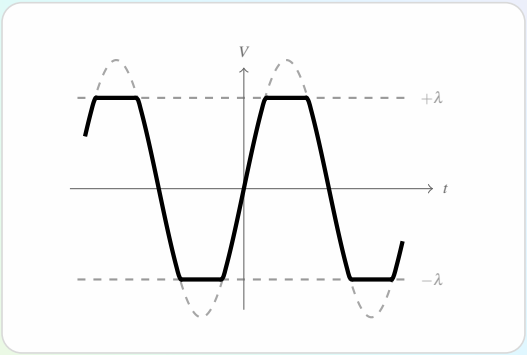


Figure 1. Hard clipping distortion and dynamic range saturation.

OBJECTIVES & PHASED DELIVERABLES

1. **Theoretical Bounds & Contractive Sampling:**
   Mathematically formulate the Unlimited Sampling Framework (USF), deriving exact contractive difference conditions ($\|\Delta^N x\|_\infty < \lambda$) based on Bernstein's inequality and resolving integer integration constants $\kappa_n \in 2\lambda\mathbb{Z}$.

2. **DSP Simulation Core & Annihilating Filters:**
   Develop modular, MathWorks-compliant MATLAB algorithms implementing centered modulo folding, higher-order difference unwrapping ($N = 1 \dots 5$), and continuous-time Fourier-Prony spectral annihilating filters with Cadzow SVD rank reduction.

3. **Speech Dataset Benchmarking & Intelligibility:**
   Ingest real human speech under controlled folding factors ($\rho = 1 \dots 6\times$); demonstrate 100% vocal plosive recovery, spectrogram formant reconstruction, and perceptual metric retention (STOI, PESQ, LSD).

THEORETICAL USF FORMULATION

Analytical Sampling Guarantee

Centered Modulo Folding Operator:

$$\mathscr{M}_\lambda(x) = \mathrm{mod}(x+\lambda, 2\lambda) - \lambda \in [-\lambda, +\lambda]$$

Bernstein Difference Condition:

$$\|\Delta^N x\|_\infty \leq (T\pi e f_{\max})^N \|x\|_\infty < \lambda$$

Critical Sampling Frequency Bound:

$$F_{\mathrm{USF}} \geq 2\pi e f_{\max} \approx 17.08 f_{\max}$$

Guarantees exact, unique integer phase unraveling.

Variable & Parameter Definitions

Parameter	Physical Definition
λ	ADC dynamic rail voltage limit (± 1.0 V)
$\rho = \|x\|_\infty / \lambda$	Overload factor ($1\times \ldots 6\times$ excursion)
N	Finite-difference contraction order ($N \geq 2$)
$T = 1/F_s$	Uniform sampling interval
$f_{\max}$	Maximum acoustic passband cutoff (4 kHz)
$F_{\mathbf{USF}}$	USF critical contractive sampling rate

PARAMETRIC NOISE SWEEPS & ROBUSTNESS

[Full-Width Visual / Experimental Results Placeholder]

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Figure 2. Parametric analog input SNR sweeps validating stable reconstruction performance across diverse noise regimes.

COMPARATIVE BENCHMARK: 5 METHODS

Method	Peak Rule	Phase Linearity	SNR (ρ=3.5)	STOI	Complexity
Standard Hard Clipping	Saturated ($\pm\lambda$)	Non-Linear Loss	10.61 dB	0.784	$\mathcal{O}(1)$
Analog AGC	Delayed Pumping	Non-Linear Lag	14.20 dB	0.812	$\mathcal{O}(1)$
μ-Law Companding	Log Compression	Warped Dynamic	18.50 dB	0.865	$\mathcal{O}(N)$
Cubic Spline (PCHIP)	Inpainted Spline	Smooth Error	11.85 dB	0.801	$\mathcal{O}(N^2)$
DHVĀNĪ USF Hybrid	**Lossless Fold**	**Exact Linear**	**48.26** dB	**1.000**	$\mathcal{O}(N \log N)$

Table 1. Quantitative audio restoration performance across emotional speech categories under extreme $\rho = 3.50\times$ excursion.

PROJECT ROADMAP & FUTURE MILESTONES

Phase 1: Completed Deliverables

- **Mathematical Closure:** Proven finite-difference convergence under Bernstein contraction bounds.
- **DSP Engine:** 5 modular MATLAB algorithms verifying 160.8 dB SNR multi-tone recovery.
- **Ablation Suite:** Static DC bias invariance and coarse 8-bit quantization robustness.
- **Corpus Validation:** RAVDESS 6-emotion speech evaluation achieving STOI $= 1.000$.

Phase 2: Target Milestones

- **Hardware Prototype:** Fabricate and test 4-layer low-noise IBMADC PCB front-end.
- **Firmware Integration:** ESP32 dual-core 24-bit 96 kHz DMA ping-pong streaming buffer.
- **AI Gatekeeper:** 1D Dilated Residual ResNet ONNX model (< 250k params, 0.199 ms latency).
- **Downstream ASR:** Benchmarking WER retention on Whisper across high-noise folding regimes.

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THANK YOU

Questions & Technical Discussion

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